

CE 333 Statics - Test 2 (A)

Problem #1 (10 points)

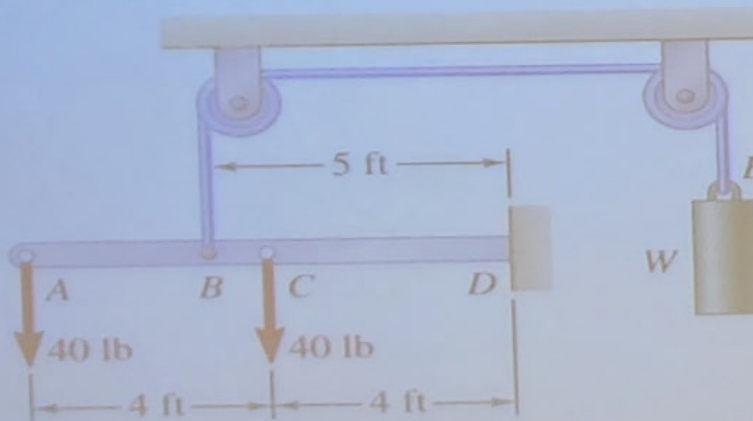
For the beam shown, determine the reaction at D (fixed-end support) when $W = 100 \text{ lb}$.

INSTRUCTIONS:

Please thoroughly read each problem before solving it. You will be graded on completion of each problem...not correctness. Make sure you look at each problem and attempt each one

Problem #1

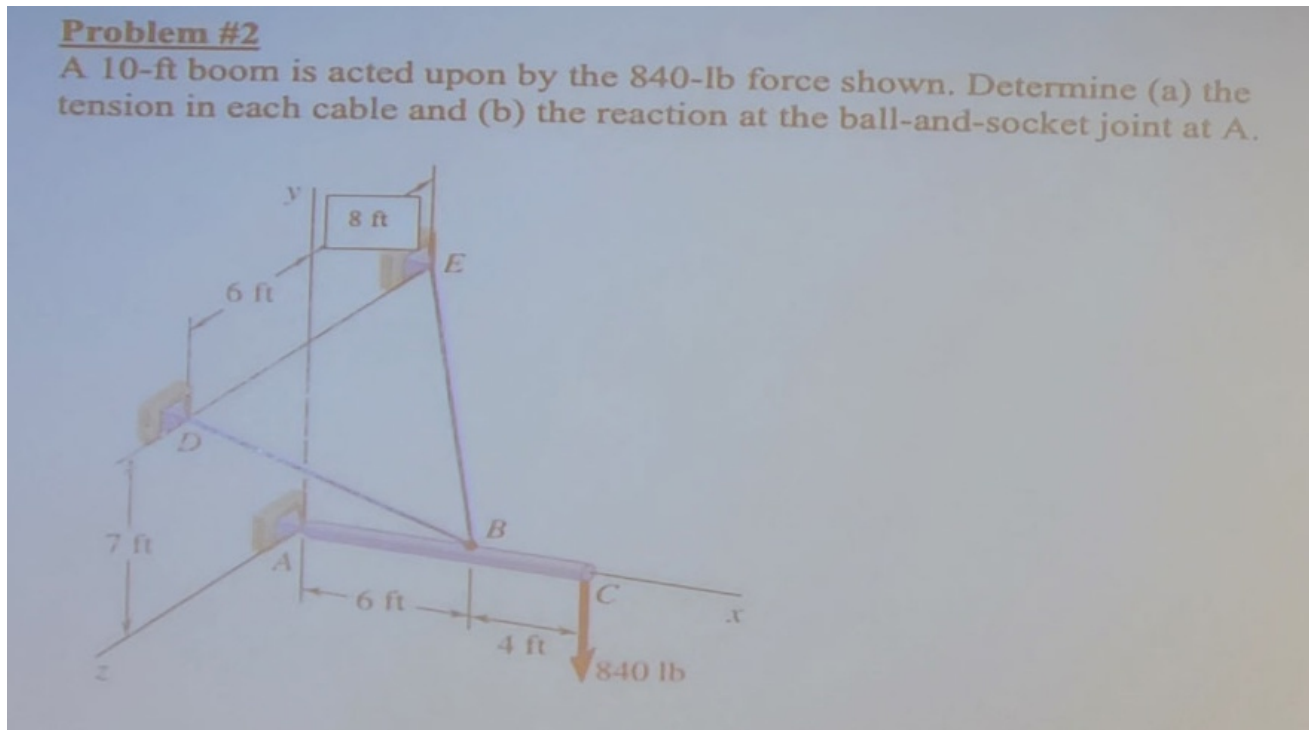
For the beam shown, determine the reaction at D (fixed-end support) when $W = 100 \text{ lb}$.



Problem #2 (15 points)

A 10-ft boom is acted upon by the 840 lb force shown. Determine

- (a) the tension in each cable and
- (b) the reaction at the ball-and-socket joint at A.



Problem #3 (12 points)

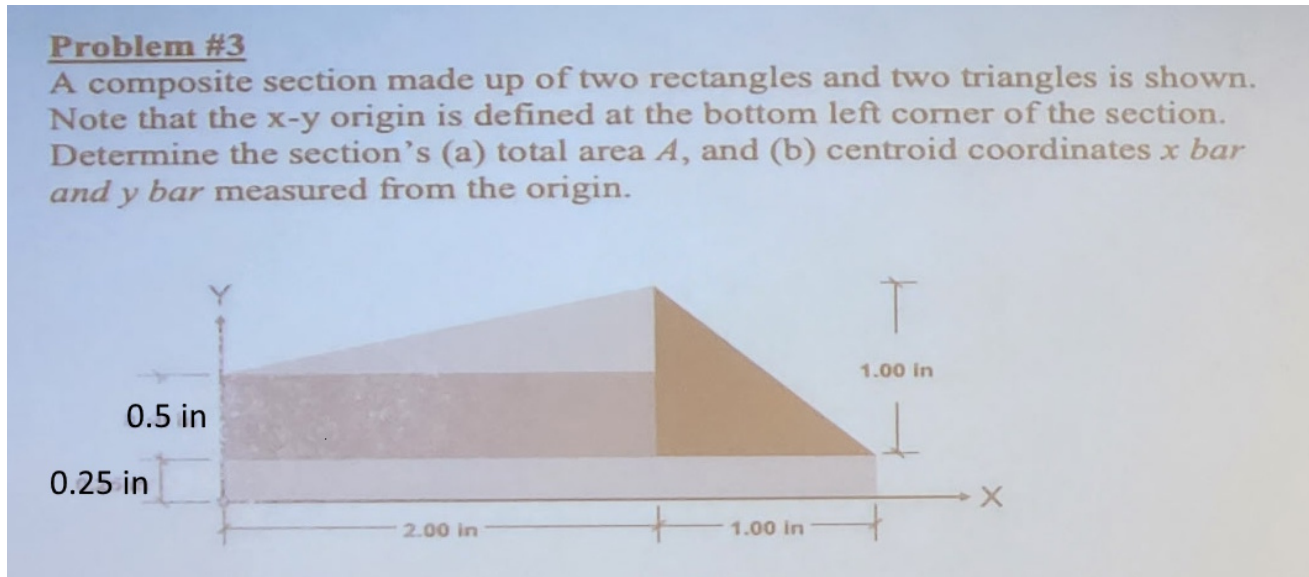
A composite section made up of two rectangles and two triangles is shown.

The x-y origin is defined at the bottom-left corner of the section.

Determine the section's

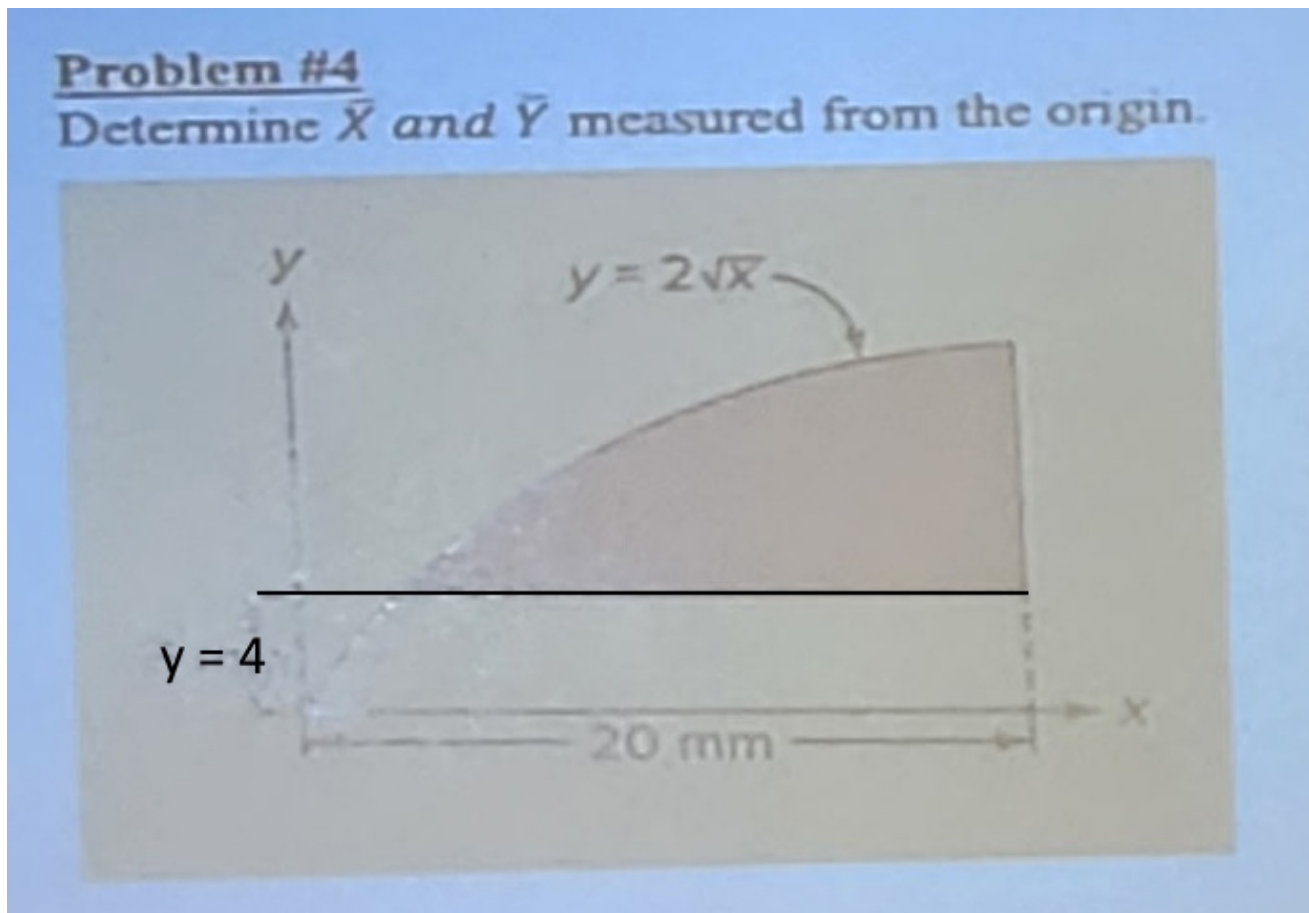
(a) total area A , and

(b) centroid coordinates $\bar{x}$ and $\bar{y}$ measured from the origin.



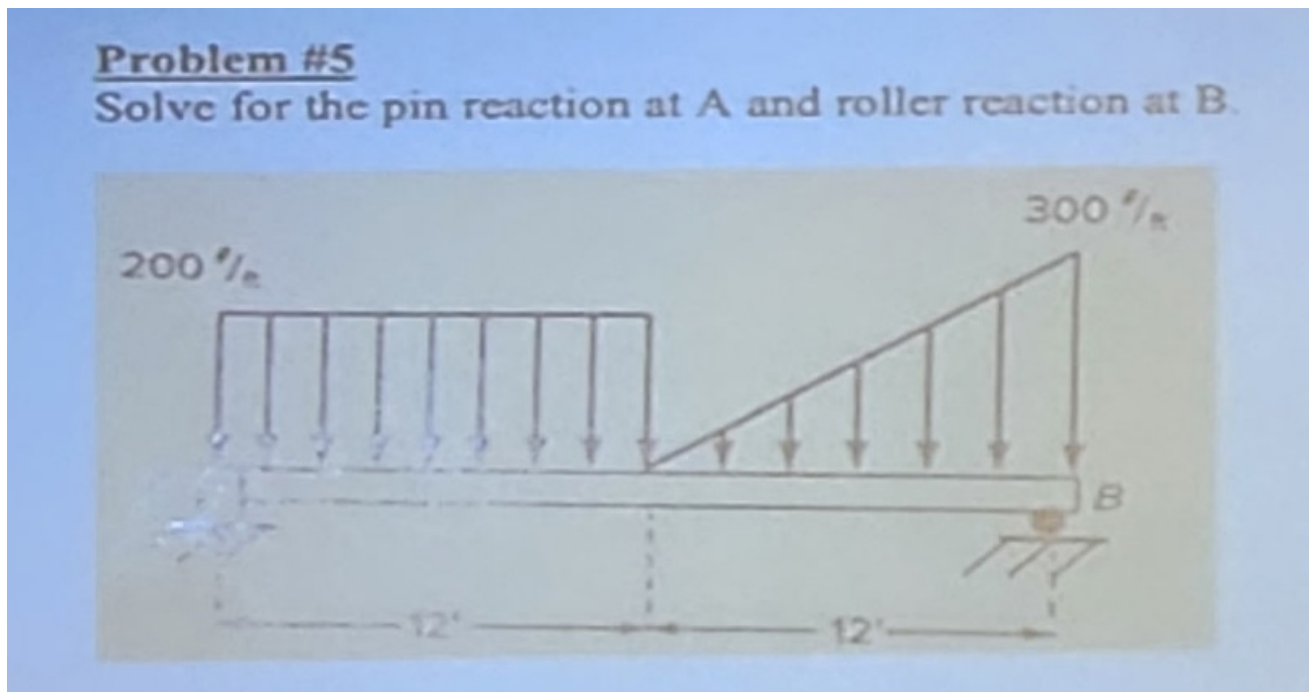
Problem #4 (15 points)

Determine $\bar{X}$ and $\bar{Y}$ measured from the origin for the area bounded by $y = 2\sqrt{x}$ and $y = 4$, with x ranging from 0 to 20 mm.



Problem #5 (13 points)

Solve for the pin reaction at A and the roller reaction at B for the beam subjected to the distributed loads shown.



Problem #6 (15 points)

Use the method of joints to solve for the force in member FE, and then use the method of sections to determine the force in member BC.

Problem #6

Use the method of joints to solve for the force in member FE, and then use the method of sections to determine the force in member BC.

